

UNIVERSITY OF CLAUDE · FREE RESOURCE MAP

Resource Map

CSCE 627 Theory of Computability — module-by-module mapping onto free university courseware

HOW TO USE THIS

This is your lecture hall. For each module: watch or read the mapped external material first, then work the module slides and notes, which are written to assume you have seen it.

Where a specific lecture number is given, that is the one that covers this module. Where a book chapter is given, read that chapter, not the whole book.

Everything here is free. Resources remain under their own licences — link to them, do not redistribute them.

Primary sources for the whole course

Resource	Role in this course
Sipser — Introduction to the Theory of Computation, 3rd edition	The primary source and the best-written textbook in theoretical computer science. Chapters 1–6 are this course. Library copy; the first chapter is free from the publisher and MIT 18.404 follows it exactly.
MIT 18.404J — Theory of Computation (free lectures and problem sets, taught by Sipser)	The author lecturing from his own book, free in full — video, notes, problem sets, and exams with solutions. This is the single best free resource in the program.
Hopcroft, Motwani & Ullman — Introduction to Automata Theory, Languages, and Computation	Stronger on Modules 02–03 than Sipser, and the parsing material connects directly to CSCE 605. Library copy.
Soare — Turing Computability; and Rogers — Theory of Recursive Functions	Module 07 and Module 09's reference — the arithmetic hierarchy and degrees, done properly. For the student who wants to go further.
Nagel & Newman — Gödel's Proof; and Smith — An Introduction to Gödel's Theorems	Module 10. Smith's book has a free study guide at this link, and it is the right level for someone who has just done Module 05.
Cousot & Cousot — Abstract Interpretation (free papers); and the Infer and Astrée documentation	Module 12's material — how production analysers live with undecidability, from the people who formalised the approach.

Module-by-module map

Module 01 — What a Computation Is

Resource	What it covers for this module
Sipser — Introduction to the Theory of Computation, chapter 0 and section 3.3	The vocabulary of §1 and the thesis discussion of §2 , stated carefully.
MIT 18.404J — lecture 1 (free video and notes)	Sipser lecturing on this material , which is the best available introduction and is free.
Turing — On Computable Numbers (1936) (free)	The original . Section 9's argument that the machine captures what a human computer does is the thesis's actual justification, and it is readable.
Matiyasevich — Hilbert's Tenth Problem (free chapters)	§1's motivating problem, resolved — and a good illustration of how long a reduction can take to construct.

Module 02 — Finite Memory

Resource	What it covers for this module
Sipser — chapter 1	Finite automata, the subset construction, and the pumping lemma — the reference for this module.
MIT 18.404J — lectures 1–3 (free)	The same material lectured , with problem sets and solutions.
Cox — Regular Expression Matching Can Be Simple And Fast (free)	§1's engine distinction, in practice — Thompson simulation against backtracking, with measurements. Required reading for anyone who uses regular expressions.
Hopcroft, Motwani & Ullman — chapters 2–4	Myhill–Nerode and DFA minimisation , treated more thoroughly than Sipser does.

Module 03 — Memory as a Stack

Resource	What it covers for this module
Sipser — chapter 2	Grammars, pushdown automata, their equivalence, and the context-free pumping lemma — the reference for §1–§3.
Hopcroft, Motwani & Ullman — chapters 5–7	Stronger on the parsing classes of §4 , including the LR(1) characterisation.
Ford — Parsing Expression Grammars (free)	§4's PEG row , including why the class is incomparable with the context-free languages.
Grune & Jacobs — Parsing Techniques, 2nd edition (first edition free)	The comprehensive survey of §4 , with the first edition freely available from the author.

Module 04 — Turing Machines

Resource	What it covers for this module
Sipser — chapter 3	The machine, the variants of §1, and the decidable/recognisable distinction of §2.
Turing — On Computable Numbers, sections 1–7 and 9 (free)	The model and the universal machine of §3, in the original , plus the section 9 argument that justifies the whole enterprise.
Minsky — Computation: Finite and Infinite Machines	§1's two-counter result , and the clearest treatment of the minimal models. Library copy.
MIT 18.404J — lectures 5–7 (free)	This module lectured , with the encoding discussion of §3 made careful.

Module 05 — Undecidability

Resource	What it covers for this module
Sipser — chapter 4	The diagonalisation argument and the undecidability of A_{TM} , with the countability preliminaries of Module 04 §4.
MIT 18.404J — lectures 8–9 (free)	Sipser proving this himself , which is worth watching even if you have read the chapter.
Turing — On Computable Numbers, section 8 (free)	The original undecidability argument , framed in terms of circle-free machines rather than halting.
Hofstadter — Gödel, Escher, Bach	§3's method, explored at length and informally — not a textbook, and a good companion to Module 10.

Module 06 — Reductions

Resource	What it covers for this module
Sipser — chapter 5	Mapping reductions, Rice's theorem, and the Post correspondence problem — the reference for this module, including the PCP reduction in full.
MIT 18.404J — lectures 10–12 (free)	The reductions worked at the board , which is the right way to see them the first time.
Rice — Classes of recursively enumerable sets (free via JSTOR listing)	§2's theorem, in the original . Short.
Berger — The undecidability of the domino problem; and the Wang tiles literature	§3's tiling result , and its connection to aperiodic tilings is worth following.

Module 07 — Degrees of Unsolvability

Resource	What it covers for this module
Sipser — sections 6.3–6.4	Oracles, Turing reducibility, and the hierarchy — a brief but sufficient treatment.

Resource	What it covers for this module
Soare — Turing Computability: Theory and Applications	The modern reference for §4 , including the priority method and Post's problem.
Rogers — Theory of Recursive Functions and Effective Computability	The classical reference , and still the clearest treatment of the arithmetic hierarchy of §2. Library copy.
Baier & Katoen — Principles of Model Checking	§3's safety/liveness distinction, operationalised — the two different algorithms and why. Library copy; the lecture notes from the authors' courses are free.

Module 08 — Undecidability in the Wild

Resource	What it covers for this module
Landi — Undecidability of static analysis (free)	§1's aliasing result, proved — may-alias is undecidable and so is may-alias for a restricted language.
Pierce — Types and Programming Languages	§2's decidability boundaries — which type systems are decidable and why the restrictions are where they are. Library copy; the exercises are free.
Cohen — Computer Viruses: Theory and Experiments (free)	§3's virus-detection undecidability , in the paper that introduced the term.
Clarke, Biere, Raimi & Zhu — Bounded Model Checking (free)	§3's method — and the clearest statement of what a bounded result does and does not establish.

Module 09 — Computable Functions

Resource	What it covers for this module
Sipser — sections 6.1–6.2 and 6.4	The recursion theorem of §3 and an introduction to §4's Kolmogorov complexity.
Li & Vitányi — An Introduction to Kolmogorov Complexity and Its Applications	The reference for §4 , including the randomness definition and its applications. Library copy.
Aaronson — The Busy Beaver Frontier (free)	§4's busy beaver material , including the independence results and the state of knowledge. Unusually readable.
The bbchallenge collaboration (free)	The BB(5) proof, completed in 2024 , with the whole effort documented — a good example of what settling one value of an uncomputable function actually requires.

Module 10 — Logic and Incompleteness

Resource	What it covers for this module
Sipser — section 6.2	§2's derivation of incompleteness from undecidability , which is the route this module takes.
Smith — An Introduction to Gödel's Theorems (free study guide)	The careful treatment of §3 , with the conditions stated properly. The author's free Teach Yourself Logic guide is also excellent.
Nagel & Newman — Gödel's Proof	The short classic account of §2's original route , including arithmetisation. Library copy, and under 150 pages.
Franzén — Gödel's Theorem: An Incomplete Guide to Its Use and Abuse	§3's callout, at book length — the standard correction of the misreadings, and worth reading before citing the theorems anywhere.

Module 11 — Other Models

Resource	What it covers for this module
Barendregt — The Lambda Calculus; and Pierce's TAPL chapter 5	§1's calculus , with Pierce's treatment being the right entry point for a programmer.
Cook — Universality in Elementary Cellular Automata (free)	§2's Rule 110 result , which is remarkable and surprisingly readable.
Churchill, Biderman & Herrick — Magic: The Gathering is Turing Complete (free)	§2's last row , done properly — and a good illustration of how little a system needs.
Aaronson — NP-complete Problems and Physical Reality (free)	§3 and §4 assessed carefully — what physics does and does not offer computation, including every proposal in §4.

Module 12 — The Engineering Response

Resource	What it covers for this module
Cousot & Cousot — Abstract Interpretation papers (free)	§1's framework from its originators , including the widening operator and the soundness-by-construction argument.
Blanchet et al. — A Static Analyzer for Large Safety-Critical Software (Astrée) (free)	§1's industrial result , including what the domain tuning actually involved.
Necula & Lee — Proof-Carrying Code (free)	§3's approach , and the clearest statement of the check-versus-find asymmetry.
Facebook Infer — the design documentation (free)	§4's near-complete position, stated explicitly by its authors — a rare and commendable piece of honest tool documentation.

Module 13 — The Limits, Honestly

Resource	What it covers for this module
Franzén — Gödel's Theorem: An Incomplete Guide to Its Use and Abuse	§2's second and sixth rows, at book length — the standard correction, and worth reading before citing the theorems.
Aaronson — Why Philosophers Should Care About Computational Complexity (free)	The honest version of what these results do and do not bear on , spanning this course and CSCE 637.
Sipser — the whole book, reread	It is short, and the second reading is substantially different from the first — which is unusual for a textbook and is worth the weekend.
Rice's theorem, Module 06, reread	If one result from this course should be at your fingertips, it is this one — it answers 'is this analysis possible?' in ten minutes, repeatedly, for the rest of your career.